

NIGERIA ELECTION PERFORMANCE INTELLIGENCE DASHBOARD

PROJECT REPORT

Interactive Election Analytics Dashboard
Microsoft Excel & VBA

Prepared by
Wahab Sodiq

Project Overview

The Nigeria Election Performance Intelligence Dashboard is an interactive data analytics project designed to analyze and visualize election performance across Nigeria.

The project transforms election data into an interactive decision-support dashboard that enables users to explore election outcomes, voter participation, candidate performance, political party performance, and geographical trends across different election years and locations.

The dashboard covers:

- Election Years: 2007–2023
- Election Types: Presidential, Governorship, Senatorial, and House of Representatives
- Geographical Coverage: 36 States + FCT
- Local Government Areas: 774 LGAs

Note: The dataset used for this project is synthetic and was created for analytical and portfolio purposes. The results, candidates, and winners displayed should not be interpreted as official historical election results.

Business Problem

Election data can contain large volumes of information across multiple years, election types, political parties, candidates, states, and Local Government Areas. Analyzing this information manually makes it difficult to quickly answer important questions such as:

- Which election recorded the highest voter participation?
- Which states had the highest number of accredited voters?
- Who won a particular election?
- Which political party performed best?
- How did candidate performance vary across elections?
- How does voter turnout differ across states and LGAs?
- How do election metrics change across different election years?

The goal of this project was to build an interactive dashboard that transforms raw election data into meaningful insights that can be easily explored by users.

Project Objectives

The major objectives of this project were to:

- Analyze election performance across multiple election years.
- Monitor voter participation and turnout.
- Compare registered voters, accredited voters, valid votes, and rejected votes.
- Identify election winners and winning political parties.
- Analyze candidate performance.
- Evaluate political party performance.

- Analyze election results across States and LGAs.
- Provide geographical visualization of election participation across Nigeria.
- Build an interactive dashboard with dynamic filtering and navigation.
- Create a user-friendly analytical experience using Microsoft Excel.

Tools & Technologies Used

Microsoft Excel was used for data cleaning, data transformation, data validation, Pivot Tables, Pivot Charts, slicers, advanced formulas, named ranges, helper cells, Form Control Combo Boxes, and dashboard design.

VBA (Visual Basic for Applications) was used to enhance dashboard interactivity and functionality, including:

- Connecting Combo Box selections to slicers.
- Managing Election Year, Election Type, Zone, State, LGA, and Party filters.
- Creating dependent State → LGA filtering.
- Improving filter response speed.
- Updating winner images dynamically.
- Updating State/LGA winner visuals.
- Updating House of Representatives winner visuals.
- Updating dynamic charts.

Dataset Overview

The dashboard analyzes election-related information across Nigeria.

The data includes important fields such as:

- Election Year
- Election Type
- Zone
- State
- Local Government Area
- Political Party
- Candidate
- Registered Voters
- Accredited Voters
- Valid Votes
- Rejected Votes
- Total Votes
- Winning Candidate
- Winning Party
- Vote Share
- Winning Margin

The dataset supports analysis at both national and geographical levels.

Data Preparation

Before building the dashboard, the raw data was prepared for analysis.

1. Data Cleaning

The dataset was reviewed to identify and address inconsistent values, incorrect data formats, blank values, duplicate records, and formatting inconsistencies.

2. Data Structuring

The cleaned dataset was organized into structured Excel tables to support Pivot Tables, dynamic formulas, named ranges, and dashboard calculations.

3. Data Modeling

Helper tables and summary tables were created to support dashboard calculations and interactive visuals, including election winners, winning parties, vote shares, winning margins, voter turnout, state performance, candidate performance, and party performance.

Dashboard Structure

The project was designed as a complete analytical dashboard system consisting of multiple pages:

1. Cover Page

Serves as the entry point into the dashboard and provides navigation to the main analytical experience.

2. Executive Overview

Provides a high-level summary of election performance across Nigeria. Key Performance Indicators include Registered Voters, Accredited Voters, Overall Turnout, Valid Votes, Rejected Votes, and Rejection Rate.

Key visualizations include:

- Accredited Voters by Election Year
- Accredited Voters by State
- Total Accredited Voters by Election Type
- Presidential Winner Card
- Executive Insights

The Presidential Winner Card dynamically displays the winner image, winning party, winning votes, winning margin, and winning vote share based on the selected Election Year and Election Type.

3. Election Analysis

Provides deeper examination of election performance across election years, election types, political parties, and candidates.

4. State & LGA Analysis

Focuses on geographical election analysis at State and Local Government Area levels. Features include State-level analysis, LGA-level analysis, State vs National Turnout comparison, dynamic geographical filtering, dependent State → LGA filtering, and State/LGA winner analysis.

5. Candidate Analysis

Focuses on candidate votes, vote share, ranking, and performance across elections.

6. Party Analysis

Provides insights into party vote performance, participation, rankings, and winning performance.

7. Data Dictionary

Explains important fields and metrics used throughout the dataset and dashboard.

8. About Dashboard

Provides information about the project purpose, functionality, dataset coverage, analytical objectives, and navigation instructions.

Interactive Filtering System

The dashboard uses six major interactive filters:

- Election Type
- Election Year
- Zone
- State
- LGA
- Party

The filters are implemented using Form Control Combo Boxes connected to VBA-powered slicers. This approach provides a cleaner and more professional dashboard interface while preserving interactive filtering capabilities.

A dependent filtering system was implemented so that when a user selects a State, the LGA Combo Box updates to display only LGAs belonging to that State. This improves usability and prevents irrelevant selections.

Performance Optimization

One of the major technical challenges encountered during the project was dashboard response speed, particularly when filtering LGAs. Initially, selecting an LGA could take several minutes to respond.

The issue was resolved by optimizing the VBA filtering process through:

- Disabling Screen Updating during processing.
- Temporarily disabling Events.
- Controlling Calculation Mode when necessary.
- Using `PivotTable.ManualUpdate`.
- Updating only Pivot Tables connected to the relevant slicer.
- Avoiding unnecessary `RefreshAll` operations.
- Using `VisibleSlicerItemsList` for faster slicer filtering.

Result: The LGA filter response was significantly improved and became almost instantaneous.

Dynamic Executive Insights

The Executive Overview includes automatically generated insights based on selected election filters.

The insights dynamically communicate:

- Election winner and winning vote share.
- Number of accredited voters and voter turnout.
- The geographical area with the highest accredited voter participation.
- How decisively an election was won through winning margin analysis.

The insight system was built using INDEX, MATCH, MAX, IFERROR, Pivot Table outputs, and Helper Cells.

Geographical Analysis

The dashboard includes a geographical visualization of Nigeria to help users understand how accredited voter participation varies across States.

Users can dynamically explore geographical election performance using the dashboard filters.

Dynamic Winner Images

A major interactive feature of the project is the use of dynamically changing winner images.

VBA was used to:

1. Identify the selected Election Year and Election Type.
2. Generate a unique election key.
3. Match the key to the appropriate winner image.
4. Display the image dynamically on the Winner Card.

The VBA logic was also improved to ensure users remain on their current dashboard page instead of being automatically redirected to the Executive Overview.

Major Technical Challenges Solved

Challenge 1: Dashboard Redirection

Problem: Changing filters on other pages redirected users back to the Executive Overview.

Solution: The VBA macro was updated to store the user's currently active worksheet before updating winner images and restore that worksheet after processing.

Challenge 2: Winner Card Showing Default Results

Problem: The State & LGA Winner Card displayed default results when not all required filters were selected.

Solution: Conditional formulas were added to ensure the card remains blank until the required Election Year, Election Type, and State conditions are satisfied.

Challenge 3: State vs National Turnout Evaluating Too Early

Problem: The chart responded when only a State was selected.

Solution: Additional formula conditions were introduced to ensure the chart remains blank until the required Election Year and Election Type are selected.

Challenge 4: Slow LGA Filtering

Problem: LGA filtering initially took several minutes.

Solution: The VBA filtering logic was optimized using manual Pivot updates, faster slicer selection, reduced workbook refreshing, and controlled calculation settings.

Challenge 5: Pivot Table Overwrite Warning

Problem: Changing filters generated a Pivot Table overwrite warning.

Solution: The Pivot Table layout and refresh behavior were corrected to prevent conflicting expansion areas.

Challenge 6: Returning to the Cover Page

Problem: Users could navigate through the dashboard but could not easily return to the Cover Page.

Solution: The project logo and title area were grouped and used as a consistent navigation button across dashboard pages.

Key Insights Enabled by the Dashboard

The dashboard enables users to quickly identify:

- Election participation trends across years.
- States with high voter participation.
- Election types with the highest accredited voters.
- Election winners and winning parties.
- Candidate performance.
- Political party performance.
- Voter turnout patterns.
- Winning margins.
- State and LGA election performance.

Dashboard Design Approach

The dashboard was designed around the following principles:

User-Friendly Navigation

Users can move easily between different analytical pages.

Consistent Branding

A consistent Nigeria election-themed design was maintained throughout the project.

Interactive Analysis

Users can dynamically filter the dashboard without manually manipulating Pivot Tables.

Executive-Level Reporting

The Executive Overview provides important insights at a glance.

Minimal Scrolling

The dashboard was designed as a polished analytical interface rather than a long scrolling report.

Skills Demonstrated

This project demonstrates practical knowledge in:

Microsoft Excel

- Advanced formulas
- Pivot Tables
- Pivot Charts
- Named Ranges
- Conditional Logic
- Data Cleaning

- Data Transformation

Dashboard Development

- KPI Development
- Interactive Dashboard Design
- Data Visualization
- UX/UI Design
- Navigation Systems
- Executive Reporting

VBA

- Automation
- Slicer Management
- Pivot Table Optimization
- Dynamic Filtering
- Worksheet Navigation
- Shape Manipulation
- Dynamic Image Updates
- Performance Optimization

Key Takeaways

Building this project provided practical experience beyond simply creating charts.

The project involved solving real dashboard development challenges, including:

- Performance optimization.
- Complex filter interactions.
- Dependent filtering.
- Dynamic calculations.
- Pivot Table behavior.
- VBA automation.
- Dashboard navigation.
- User experience design.

This project strengthened my understanding of how different Excel features can work together to create a complete interactive business intelligence solution.

Future Improvements

Possible future improvements include:

- Rebuilding the project in Power BI.
- Adding DAX measures for more advanced calculations.
- Creating drill-through analysis.
- Adding tooltips and deeper contextual insights.
- Connecting the dashboard to a live or regularly updated data source.
- Adding advanced trend analysis.
- Introducing predictive analytics for voter participation trends.

Project Author

Wahab Sodiq

Aspiring / Developing Data Analyst

Skills Applied:

Microsoft Excel | Data Analysis | Data Visualization | Data Cleaning | VBA Automation | Dashboard Development